

HAT-P-32b

R-0252

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_230914 · created 2026-07-17T06:44:33+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460201.8207 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1687 +/- 0.0053
Transit depth (Rp/Rs)^2	2.85 +/- 0.18 [%]
Orbital Inclination (inc)	86.38 +/- 1.36
Airmass coefficient 1 (a1)	1.1849 +/- 0.0082
Airmass coefficient 2 (a2)	-0.162 +/- 0.013
Scatter in the residuals of the lightcurve fit is	0.7 %
Best Comparison Star	None
Optimal Aperture	5.02
Optimal Annulus	7.72
Transit Duration (day)	0.1263 +/- 0.0028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1687 ± 0.0053 vs 0.1489 (deviation 3.74σ)
Duration fit vs published	0.1263 d vs 0.1304 d (deviation 1.47σ)
Mid-time O–C	2.6 min
Depth SNR	31.8
Residual scatter	0.7 %

Light curve

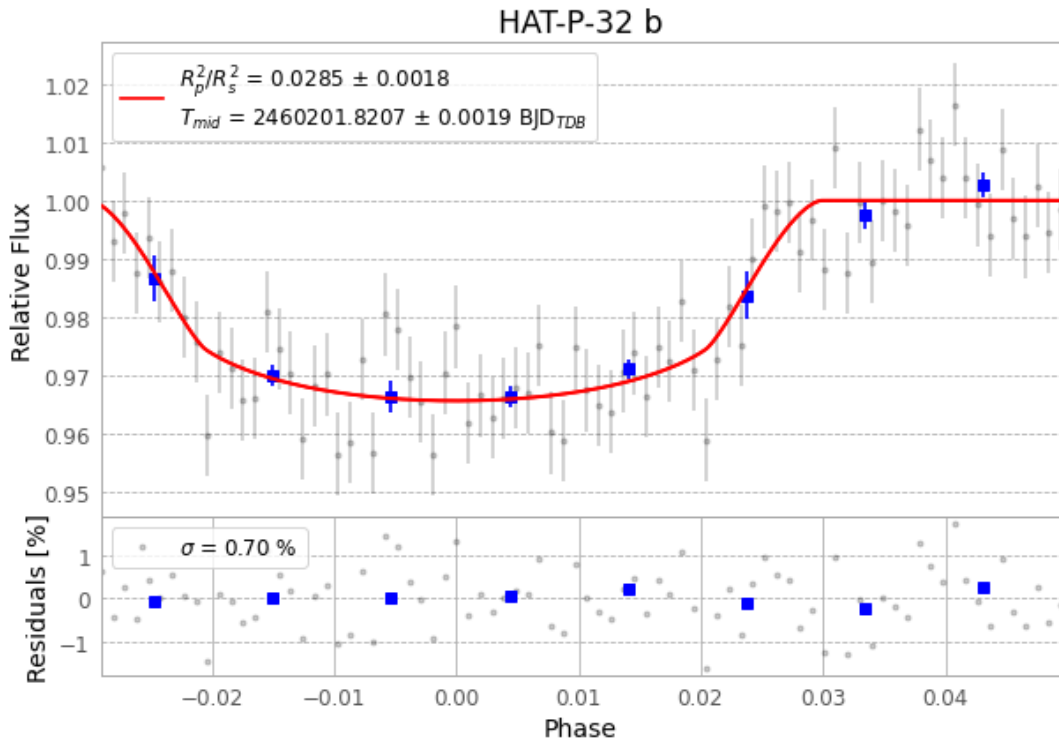


Figure 1 — FinalLightCurve_HAT-P-32 b_2023-09-13.png (SHA-256 1156b9bf0a2b243c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [261, 253], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 25e66de1e4bfe178...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

82 FITS frames (54 MB), HATP-32230914060615.FITS → HATP-32230914100915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-230914.log (db0ba3a35be5...), FinalLightCurve_HAT-P-32 b_2023-09-13.png (1156b9bf0a2b...), FinalLightCurve_HAT-P-32 b_2023-09-13.csv (964b48e94721...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 06:44Z.